

Derek Peng

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EDUCATION

University of California, Berkeley

Expected Graduation: May 2028

B.S. Electrical Engineering & Computer Science

GPA: 3.87

Awards & Honors: Berkeley Regents Scholar (1%), RoboSub Semi-Finalist, CyberPatriot National Finalist

WORK EXPERIENCE

Doppel | San Francisco, CA

May 2026 – Present

Software Engineering Intern

- Shipped alert routing analysis platform to 150+ enterprises, eliminating 300+ hours of triage per month
- Rebuilt DNS tooling, recovering ~3,500 falsely-resolved alerts yearly for Robinhood, Coinbase, OpenAI
- Building agentic LLM system to discover novel threat signals and trigger live detections without analysts

AI4Sec Lab | Rochester, NY

April 2025 – Present

Research Intern under Professor Rastogi

- Pioneered attack-agnostic defense mechanisms that classify tampered-with images with 90%+ accuracy
- Reduced ImageNette misclassification by 57.7% over state-of-the-art by leveraging concept attribution
- Accepted and presenting to academia and industry at NeurIPS 2025: Reliable ML From Unreliable Data

Cal Poly Pomona Security Lab | Pomona, CA

Aug 2024 – December 2024

Research Intern under Professor Husain

- Published as lead author for IEEE paper on malware classification using Vision Transformer architecture
- Achieved a 12% accuracy improvement and 30% reduction in training time compared to CNN baselines
- Presented on technical paper and novel findings to 100+ faculty and cybersecurity researchers at MIT

Dasion | Claremont, CA

May 2023 – Aug 2023

Robotics & Computer Science Intern

- Conducted a market analysis with Python for robotics for healthcare market to identify market gaps
- Modeled MVP and gave business pitch for a caretaker robot based on 50+ surveys and feedback forms
- Utilized Robot Operating System and Gazebo to create virtual testing environment to demo for engineers

Peraton | Monrovia, CA

June 2024 – Aug 2024

Engineering Intern

- Created and presented cybersecurity safety material to 100+ employees in the Deep Space Network team
- Researched and created hosted database of military cybersecurity standards through STIGs and NIST-800
- Worked with mentors to integrate cybersecurity through DevSecOps into the workflow of the company

ACTIVITIES

Phantom | Berkeley, CA

Jan 2026 - May 2026

Technical Consultant

- Leading research on Gen Z behavior/adoption of Phantom's Kalshi/prediction-market integration
- Creating an incentive-aligned prediction-market game where players keep profits to show trading behavior
- Tracking all trades and bet justifications to analyze consumer risk-taking, topic choices, and social drivers

Acorns | Berkeley, CA

Aug 2025 - Dec 2025

Technical Consultant

- Directed Go To Market Strategy for new service-based industry micro-investing tipping feature
- Surveyed 120+ barbershops nationwide on tipping preferences to best form an ideal customer profile
- Established web application MVP with Django based on iterative feedback from partnered local shops

RoboSub Competition | Fullerton, CA

Oct 2021 – May 2025

Founder & Team Captain

- Pioneered school's first-ever construction of an AUV to compete in collegiate RoboSub competition
- Leveraged ZED2 Stereo Camera with YoloV6 processing and OpenCV for autonomous object detection
- Trained and mentored 30+ new members over a six-week period, growing the team 70% over a year

Solo Engineer

- Building an institutional-grade trading card exchange using Next.js and Supabase Realtime for price feeds
- Routing \$22k+ in volume across 710+ trades with Stripe Connect for secure KYC-compliant settlement
- Engineered an AI intake pipeline using Gemini Vision and PSA's API to instantly onboard graded cards

PROJECTS

- **Robot Tour** | Constructed a fully autonomous pathfinding robot using an Arduino microcontroller to navigate an obstacle course. Used an IMU and stepper motors for precise movement and state-estimation
- **tash.cards** | Engineered an institutional-grade trading card exchange using Next.js and Supabase, routing \$22k+ in volume across 710+ trades, and integrating Stripe Connect for KYC-compliant settlement
- **Cronos Guard** | Developed a real-time cryptocurrency wallet security agent using transaction simulation and vulnerability scanning to proactively detect and block malicious smart contracts to prevent fraud
- **Signal** | Engineered a Python auditd ingestion pipeline that submits cryptographic log fingerprints to a permissioned network implemented through Hyperledger Fabric for tamper-proof validation of audit logs
- **Kairos** | Programmed a Linux hardening script in Bash that automated vulnerability patching, permission auditing, and firewall configuration, according to NIST 800-53 guidelines, to secure sandboxed systems
- **BastionDefense** | Founded a cybersecurity competition. Hosted backend with Kubernetes and organized weeklong red-blue team engagement challenges, security lessons & workshops with 50+ nationwide teams

SKILLS & INTERESTS

Technical Skills: Python, C++, Rust, Solidworks CAD, Railway, GraphQL, Vercel, Bash, Linux, QGroundControl, Robot Operating System, VMware, VSphere, Docker, Notion, Obsidian, Gitbook, Git

Interests: AI Safety and Alignment, Penetration Testing, Startups and VC, Autonomous Robotic Systems